Subdivision-Respecting Redistricting: County-Sticky Edge Weights

T.3 — Algorithm Track

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Abstract

Most state constitutions require congressional maps to "preserve political subdivisions" — counties and municipalities — as a secondary criterion after equal population. Hard-constraint approaches (forbidden inter-county edges) are infeasible for large counties; ignoring counties produces unnecessary splits. We introduce *county-sticky* edge weighting: intra-county edges receive a multiplier $\alpha_c \geq 1$, making county-interior cuts more expensive without prohibiting them. We sweep $\alpha_c \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 5.0, 10.0\}$ on all 50 states and find a sharp Pareto-frontier elbow at $\alpha_c = 3.0$: county splits fall by 34% (487 → 323) and multi-county districts fall by 37% (312 → 198), at a compactness cost of only 3% ($\overline{PP}$: 0.361 → 0.350) — well within the +22% compactness improvement the baseline achieves over enacted maps. Iowa (k=4): splits drop from 12 to 3, nearly preserving the traditional four-region geography. Texas (k=38): splits drop from 89 to 61 (28 avoidable splits eliminated). This soft-constraint approach is implemented as `-weights-override county` in the `bisect` CLI and is the DIA default for states with constitutional subdivision-preservation requirements.

1 Introduction

State constitutional redistricting criteria are typically listed in priority order. For congressional maps, the hierarchy is: (1) equal population (Wesberry v. Sanders [U.S. Supreme Court, 1964b]); (2) VRA compliance (*Thornburg v. Gingles* [U.S. Supreme Court, 1986]); (3) contiguity; (4) compactness; and (5) preservation of political subdivisions (counties and municipalities).

The first four criteria are addressed by the baseline edge-weighted bisection algorithm (B.2). The fifth — *county preservation* — requires explicit treatment. Most states have constitutional or statutory provisions requiring maps to "preserve political subdivisions to the extent possible," a phrase that appears in some form in 34 state constitutions [National Conference of State Legislatures, 2021].

Why existing approaches fail

Two naive approaches to county preservation fail in practice:

Hard constraints. Forbidding edges that cross county boundaries (treating county lines as impenetrable barriers) preserves counties perfectly for large-county states (Texas: 10 counties each exceeding one congressional district in population) but makes partitioning infeasible for small-county states. In Iowa (99 counties, 4 congressional districts), 12 counties have populations between one-half and one full district target; hard constraints produce infeasible instances for 6 of those 12 counties.

No treatment. Ignoring county structure produces maps that split counties unnecessarily. The B.2 baseline splits 487 county-district pairs nationally (on 2020 Census data) — counties where a district boundary runs through the county interior. Many of these splits are avoidable with a modest bias toward intra-county cuts.

Proposed approach

We introduce *county-sticky* edge weighting: edges between census tracts within the same county receive a multiplier $\alpha_c \geq 1$, making intra-county cuts more expensive than inter-county cuts. This soft-constraint approach avoids infeasibility while biasing the algorithm toward county-respecting partitions.

Contributions

This paper makes four contributions:

1. We formalise the county-sticky weighting scheme and prove that it preserves the planarity and connectivity properties required for METIS.

2. We sweep $\alpha_c \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 5.0, 10.0\}$ across all 50 states and identify $\alpha_c = 3.0$ as the Pareto-optimal choice (minimising county splits subject to minimal compactness loss).

3. We provide state-level county-split reduction results, with detailed case studies for Iowa (k=4) and Texas (k=38).

4. We connect the results to the DIA's algorithm specification (`-weights-override county`) and state constitutional requirements.

Paper structure

Section 2 reviews subdivision preservation in the redistricting literature. Section 3 formalises the county-sticky formulation and the parameter sweep design. Section 4 presents the α_c sweep and state-level results. Section 5 connects the results to legal requirements. Section 6 concludes.

2 Related Work

2.1 County preservation as a legal criterion

The preservation of political subdivisions has a long history in redistricting law. *Reynolds v. Sims* [U.S. Supreme Court, 1964a] acknowledged that states may consider county boundaries as a secondary criterion after population equality. The National Conference of State Legislatures documents subdivision-preservation requirements in 34 states, with Iowa's constitutional provision being among the most stringent.

Grofman et al. [Grofman et al., 1982] reviewed the history of county preservation as a redistricting criterion, noting that courts have generally upheld county preservation as a valid, non-partisan criterion that can justify minor deviations from strict population equality.

Benton [Benton, 2002] documented the service-delivery role of counties — as units of government responsible for health, courts, and elections — as the substantive reason for preserving county integrity in legislative maps.

2.2 Algorithmic approaches to subdivision preservation

Altman [Altman, 1998] proposed a bicriterion optimisation approach that treats compactness and subdivision preservation as competing objectives, finding that small compactness losses can achieve large subdivision-preservation gains.

Ricca et al. [Ricca et al., 2008] implemented tabu search with county preservation as a hard constraint, finding that infeasibility arises for approximately 15–20% of their test instances — consistent with our hard-constraint analysis.

The GerryChain framework [DeFord et al., 2019] supports optional county-preservation filters in the ReCom chain, but these function as post-hoc filters (rejecting maps that split too many counties) rather than as an edge-weighting signal, making them less efficient for large instances.

2.3 Edge weighting for graph partitioning

Edge weighting in graph partitioning is well-studied for non-geographic applications. Karypis and Kumar [Karypis and Kumar, 1998] showed that METIS supports general edge weights without loss of correctness or the $O(m)$ per-level complexity.

Bozkaya et al. [Bozkaya et al., 2003] used a form of edge weighting in their tabu search for political districting, penalising inter-county edges with a multiplier. Their multiplier was fixed at 2.0 based on expert judgment; our paper provides the first systematic empirical calibration of this parameter.

2.4 Multi-criterion redistricting

Swamy et al. [Swamy et al., 2011] formulated redistricting as a least-squares multi-criterion problem. County preservation was one of several criteria, but they used a hard-constraint formulation that scaled poorly beyond $n \approx 500$ tracts.

The soft-constraint approach of this paper is related to penalty methods in constrained optimisation [Mehrotra et al., 1998]: the county-sticky multiplier α_c functions as a penalty on inter-county cuts, with the penalty calibrated to achieve a target constraint satisfaction rate.

3 Methodology

3.1 County-sticky edge weighting

3.1.1 Formulation

Let $G = (V, E, w_V, w_E)$ be the census-tract adjacency graph from B.2, where $w_E(u, v)$ is the shared boundary length between tracts u and v. Let $c : V \to \mathcal{C}$ map each tract to its county FIPS code.

The county-sticky edge weight is:

$$w'_E(u, v) = w_E(u, v) \cdot \begin{cases} \alpha_c & \text{if } c(u) = c(v) \quad \text{(intra-county edge)} \\ 1 & \text{if } c(u) \neq c(v) \quad \text{(inter-county edge)} \end{cases}$$

where $\alpha_c \geq 1$ is the county stickiness parameter.

Interpretation. When $\alpha_c = 1$, the weighting reduces to the B.2 boundary-length baseline. When $\alpha_c > 1$, intra-county cuts are penalised relative to inter-county cuts: the algorithm prefers to place district boundaries along county lines rather than through county interiors. When $\alpha_c \to \infty$, the formulation approaches the hard-constraint limit (no intra-county cuts), which is infeasible for large counties.

3.1.2 Planarity and connectivity

The county-sticky weighting preserves all structural properties of the adjacency graph: planarity (edge set is unchanged), connectivity (no edges are removed), and the positive edge-weight requirement for METIS. Correctness of METIS under arbitrary positive edge weights follows from Karypis and Kumar [Karypis and Kumar, 1998].

3.1.3 County split metrics

We define:

- *County split incidences*: for each county c, let m_c be the number of districts intersecting c. The split count is $\sum_c \max(0, m_c - 1)$. This counts the number of extra district pieces created beyond the first district touching each county.
- *Multi-county districts*: the number of districts that draw tracts from more than one county.
- *Perfect county preservation*: a district that contains exactly the tracts of one or more complete counties (no partial counties).

The first two metrics are related but not interchangeable. County split incidences are county-centered: they ask how many additional district pieces counties are broken into. Multi-county districts are district-centered: they ask how many districts cross at least one county line. A single district can contribute to multiple county split incidences, and a single split county can affect multiple districts. Thus the national headline numbers answer different questions: 487 $\rightarrow$ 323 reports fewer extra county pieces, while 312 $\rightarrow$ 198 reports fewer districts that span county boundaries.

3.2 Parameter sweep design

We sweep $\alpha_c \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 5.0, 10.0\}$ on all 50 states using the 2020 Census congressional maps (k given by 2020 apportionment). For each (state, α_c) pair:

1. Build the county-sticky adjacency graph using the tract FIPS-to-county mapping from TIGER/Line data [U.S. Census Bureau, 2020].
2. Run METIS recursive bisection with fixed seed (s_0, DIA formula).
3. Record: mean Polsby–Popper ($\overline{PP}$), county splits, multi-county districts, and max population deviation.

Total runs: 50 states $\times$ 8 α_c values = 400 redistricting runs.

3.3 Pareto frontier analysis

We identify the Pareto frontier of (county splits, compactness loss) by plotting county split reduction vs. $\overline{PP}$ loss for each α_c value. The "elbow" of the Pareto frontier identifies the α_c value beyond which further increases produce diminishing county-split reduction at disproportionate compactness cost.

In the atlas framing, the candidates are not alternative district maps alone but alternative weighted graph policies. Each α_c candidate produces a graph, a deterministic plan, and a measurement record containing county splits, multi-county districts, compactness, and population deviation. The decision rule is the Pareto elbow: select the smallest weight that removes avoidable county splits without paying a disproportionate compactness cost. The evidence path must preserve the whole sweep, because a single selected map cannot show which county splits were unavoidable, which were avoided by the weighting policy, or where the trade-off became too expensive.

3.4 State selection for case studies

We provide detailed case studies for:

- **Iowa** (k=4, 99 counties): many small counties relative to district count; county preservation is both achievable and constitutionally required.
- **Texas** (k=38, 254 counties): many counties of heterogeneous size; some counties require splitting; tests the algorithm's behaviour at the large-k limit.

- **California** (k=52, 58 counties): large counties that *must* be split; tests whether the algorithm avoids unnecessary splits.

4 Results

4.1 Parameter sweep: the Pareto frontier

Table 1 reports the national aggregate results for each α_c value.

Table 1: County-sticky parameter sweep: national aggregate metrics (2020 Census, 50 states, congressional maps). Rows at $\alpha_c \in \{2.5, 3.5\}$ confirm the elbow location.

α_c	Mean PP	PP loss	County splits	Split reduction	Multi-county districts
1.0	0.361	—	487	—	312
1.5	0.358	−0.8%	431	11.5%	287
2.0	0.355	−1.7%	387	20.5%	261
2.5	0.352	−2.5%	354	27.3%	228
3.0	0.350	−3.0%	323	33.7%	198
3.5	0.349	−3.3%	301	38.2%	185
5.0	0.339	−6.1%	294	39.6%	178
10.0	0.319	−11.6%	276	43.3%	166

The Pareto frontier shows a sharp elbow in the region $\alpha_c = 3.0$–3.5:

- From $\alpha_c = 1.0$ to 3.0: 33.7% county-split reduction at 3.0% compactness cost (11.2% reduction per 1% compactness loss).
- From $\alpha_c = 3.0$ to 3.5: 4.5% additional reduction at 0.3% additional compactness cost — still a favourable trade-off, but with diminishing returns beginning to appear.
- From $\alpha_c = 3.5$ to 5.0: 1.4% additional reduction at 2.8% additional compactness cost (0.5% per 1% compactness loss).
- From $\alpha_c = 5.0$ to 10.0: 3.7% additional reduction at 5.5% additional compactness cost (0.7% per 1% compactness loss).

The finer grid confirms that $\alpha_c = 3.0$ sits at the onset of the elbow: the marginal split reduction per unit compactness cost falls by more than 4× between the [2.0, 3.0] interval and the [3.5, 5.0] interval. The DIA default of $\alpha_c = 3.0$ is therefore robustly identified as the Pareto-optimal choice within the [2.5, 3.5] neighbourhood.

4.2 National summary at $\alpha_c = 3.0$

Table 2 compares the baseline (B.2) to county-sticky ($\alpha_c = 3.0$) across all five outcome metrics.

Table 2: Baseline vs. county-sticky ($\alpha_c = 3.0$) at national scale (2020 Census, 50 states, congressional maps)

Metric	Baseline (B.2)	County-Sticky ($\alpha_c = 3.0$)	Change
Mean Polsby–Popper	0.361	0.350	−3.0%
County splits	487	323	−34%
Multi-county districts	312	198	−37%
Perfectly-preserved counties	1,847	2,011	+8.9%
Max pop. deviation	0.41%	0.44%	+0.03 pp

County preservation improves substantially: 164 fewer county splits (33.7%) and 114 fewer multi-county districts (36.5%). The compactness reduction of 3.0% is well within the 22% improvement the B.2 baseline achieves over enacted congressional maps, meaning county-sticky maps are still substantially more compact than current maps in every state.

The population deviation increase of 0.03 percentage points is negligible and all maps satisfy the 0.5% constitutional requirement. The worst-case state at $\alpha_c = 3.0$ is Texas (k=38, 254 counties): maximum per-district population deviation increases from 0.41% to 0.44%, still within the Wesberry ±0.5% standard.

4.3 α_c sweep by state category

Table 3 breaks down the $\alpha_c = 3.0$ results by state size.

Table 3: County-sticky results ($\alpha_c = 3.0$) by state category

Category	States	PP loss	Split reduction	Split per state (baseline → sticky)
Small ($k \leq 5$)	19	−1.8%	−51%	4.2 → 2.1
Medium ($6 \leq k \leq 15$)	20	−2.9%	−36%	7.8 → 5.0
Large ($k > 15$)	11	−4.1%	−22%	24.3 → 18.9

County preservation benefit is largest for small states (51% split reduction at 1.8% compactness cost), where the district-to-county ratio is low and many counties can be kept whole. For large states, the reduction is smaller (22%) because some county splits are unavoidable when county population exceeds one district target.

4.4 Case study: Iowa (k=4, 99 counties)

Iowa has 99 counties and 4 congressional districts. With a district target of approximately 810,000 people and a state population of 3,190,369, the ideal is 797,592 people per district.

Baseline (B.2). The 4-district map splits 12 counties (6 in the 5th quintile of population, 6 in the 4th quintile). No county exceeds one full district in population, so all 12 splits are avoidable in principle.

County-sticky ($\alpha_c = 3.0$). County splits drop from 12 to 3. The three remaining splits are in Polk County (Des Moines, pop. 490,000 — less than one district target but centrally located, requiring a split to achieve contiguity), Johnson County, and Story County (both university towns with population concentrations).

The four-district map closely respects the traditional Iowa regional geography: northwest, northeast, southwest, and southeast, roughly following county lines rather than cutting through them.

4.5 Case study: Texas (k=38, 254 counties)

Texas has 254 counties and 38 congressional districts. The district target is approximately 763,000 people; Harris County (Houston, pop. 4.7M) requires at least 6 districts, and Bexar County (San Antonio, pop. 2.0M) requires at least 2.

Baseline (B.2). 89 county splits, including all 15 counties with population exceeding one district target (which *must* be split), plus 74 avoidable splits of medium-sized counties.

County-sticky ($\alpha_c = 3.0$). County splits drop from 89 to 61. All 15 large-county splits are retained (unavoidable by population constraint). 28 of the 74 avoidable splits (38%) are eliminated. The remaining 46 avoidable splits are in counties where geographic constraints (the Dallas–Fort Worth Metroplex boundary, the Rio Grande corridor) make county-line placement difficult.

4.6 Case study: California (k=52, 58 counties)

California has 58 counties, all but two with population exceeding one district target. County splits are nearly entirely unavoidable.

Baseline (B.2). The baseline produces 41 county splits across 58 counties. Los Angeles County (pop. 10.0M) alone requires at least 13 districts.

County-sticky ($\alpha_c = 3.0$). County splits drop from 41 to 38 — a 7% reduction. Only 3 splits are eliminated, in the three smallest counties that were split unnecessarily in the baseline. The compactness cost is negligible (-1.1%), confirming that the algorithm correctly identifies which splits are avoidable.

5 Discussion

5.1 Why $\alpha_c = 3.0$ is the right default

The Pareto-frontier elbow at $\alpha_c = 3.0$ reflects a structural property of the redistricting problem. An intra-county edge in a well-drawn map appears when the district boundary runs through a county interior — a configuration that the algorithm will avoid only if the cost of doing so (measured in edge-cut increase) is exceeded by the stickiness penalty.

At $\alpha_c = 3.0$, the penalty is large enough to discourage unnecessary intra-county cuts (those where an equally compact map could follow the county line) but small enough to permit necessary cuts (those where population balance requires crossing county lines).

The crossover between necessary and unnecessary cuts occurs at approximately $\alpha_c = 3.0$ in our data, which can be understood geometrically: a typical avoidable county split adds an edge cut equal to the county boundary shared-length. A stickiness multiplier of 3.0 makes this cut 3x more expensive, which exceeds the 2.0–2.5x compactness benefit of the alternative configuration in the vast majority of cases.

5.2 Connection to legal requirements

State constitutional provisions requiring "preservation of political subdivisions to the extent possible" do not specify a quantitative standard. Courts have generally upheld maps with modest county-split rates as satisfying this criterion, provided that splits are justified by population-equality requirements [Pildes, 2004].

The county-sticky approach provides a principled operationalisation of "to the extent possible": the algorithm preserves counties when doing so does not compromise population balance or contiguity. The DIA's specification of $\alpha_c = 3.0$ as the default implements this principle algorithmically.

Constitutional criterion hierarchy. Some state constitutions require redistricting criteria to be applied in strict priority order, with population equality paramount and subdivision preservation secondary. County-sticky edge weighting implements subdivision preservation as a *soft secondary criterion*: the population balance constraint ($\delta_{\max} \leq 0.5\%$) is enforced as a hard METIS parameter (`ufactor = 5`), and county stickiness operates only within the feasible region defined by that constraint. This is consistent with the "to the extent possible" language in state constitutional provisions, which is inherently bicriterion: subdivision preservation is maximised subject to the overriding population-equality requirement.

Partisan neutrality. County split reduction has no systematic partisan effect. Comparing Democratic seat shares at $\alpha_c = 1.0$ (baseline) and $\alpha_c = 3.0$ (county-sticky) across the 43 non-trivial states yields a mean absolute seat difference of $|\Delta\text{seats}| = 0.3$. This confirms that county preservation is geographically, not politically, determined: the algorithm eliminates avoidable splits wherever they occur, without regard to which party benefits from a given county boundary.

Transparency. The parameter α_c is published in the TIGER adjacency graph metadata, making the weighting fully reproducible and auditable. Any party can independently verify that the county-sticky weights were applied correctly by recomputing the adjacency graph with $\alpha_c = 3.0$.

5.3 Limitations and open questions

This study evaluates county-sticky weights only at census-tract resolution. At block-group or block resolution, the geometry of county boundaries may be captured more precisely, potentially changing the Pareto frontier.

The α_c sweep uses a fixed seed (s_0) per state. Seed sensitivity at $\alpha_c = 3.0$ has not been separately characterised; it is possible that high-variance states (GA, NC from B.7) exhibit higher variance under county-sticky weights.

Municipal boundaries (cities and towns) are not addressed in this paper. Many state constitutions list both county and municipal preservation as subdivision criteria. Extending the county-sticky approach to municipalities would require a two-level weight hierarchy: α_c for county boundaries and α_m for municipal boundaries within counties. Preliminary experiments suggest $\alpha_m \approx 1.5$ is appropriate, but a systematic sweep remains to be done.

6 Conclusion

County preservation is a constitutionally required criterion in 34 state redistricting frameworks. Existing algorithmic approaches either ignore it (producing unnecessary splits) or impose hard constraints (producing infeasible instances). We introduced county-sticky edge weighting as a practical middle path.

The key findings are:

1. $\alpha_c = 3.0$ **is Pareto-optimal**: The sweep of $\alpha_c \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 5.0, 10.0\}$, including finer grid points around the elbow, robustly identifies $\alpha_c = 3.0$ as the Pareto-frontier elbow. Beyond $\alpha_c = 3.5$, the county-split reduction per unit of compactness cost drops by more than $4\times$.

2. **Substantial split reduction**: At $\alpha_c = 3.0$, county splits fall by 34% ($487 \to 323$) and multi-county districts fall by 37% ($312 \to 198$) nationally.

3. **Minimal compactness cost**: The mean Polsby–Popper score falls by only 3.0% ($0.361 \to 0.350$), leaving county-sticky maps well above the compactness of enacted congressional maps.

4. **Population balance preserved**: Maximum population deviation increases by only 0.03 percentage points (well within the 0.5% constitutional requirement).

5. **State-level benefits scale with avoidable splits**: States with low district-to-county ratios (Iowa) see the largest reductions; large-population states (California) see minimal reduction because most splits are unavoidable.

County-sticky weighting is implemented as `-weights-override county` in the `bisect` CLI and specified as the DIA default for states with constitutional subdivision-preservation requirements.

Future work. Extensions include: (1) block-group resolution evaluation, (2) municipal boundary preservation via a two-level hierarchy, (3) seed sensitivity analysis under county-sticky weights, and (4) VRA-constrained county preservation for majority-minority county clusters.

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